Machine Learning

Learn to teach a computer to make predictions, and to test whether it works.

Who it is for. S.4 to S.6 students and school leavers. No coding experience is needed. Basic school maths is enough.

Tools. Python, Jupyter notebooks, NumPy, pandas, matplotlib, seaborn and scikit-learn. All are free and used in industry.

What your child will build

Each learner builds a prediction project from start to finish. They choose a question, prepare the data, set a simple baseline to beat, train a model, and test it once on data it has never seen. They present it in a five-minute talk and a short written report.

All project data is practice data made for the camp, so no real person's information is used.

The eight weeks

1	**Python and a first prediction**	A computer guesses a rent from 120 homes. Notebooks, values, arithmetic and reading error messages.
2	**Python you need**	Decisions, lists, loops and functions. A hand-written rule that names flowers, then a test of how often it is right.
3	**Tables and statistics**	Tables in pandas, picking and filtering rows, averages and spread, with NumPy.
4	**Clean, chart, question**	Fixing messy data, drawing charts that say something, choosing a project question. Checkpoint one.
5	**Baselines and a line**	Why a model must beat a simple guess. Fitting a line and measuring its error.
6	**Trees and fair tests**	Decision trees, and keeping a test set locked until the end. Checkpoint two.
7	**Your project**	An approved plan, a baseline and a compared model. Modern AI assistants are shown and critiqued.
8	**Hand over**	The final test, the report, and each learner demonstrates their work to parents and guardians.

Four extra two-hour foundation labs in weeks 1 to 4 (a fourth day) give more practice with Python and maths. An optional extension follows week 8 and is not assessed.

What a class looks like

Each three-hour class has the same shape. A short recall, then a guess written on paper before anything is run, then the code is run and investigated, a 15-minute break away from the screen, a second block, and a make time where each learner builds their own version. Every class ends with a sentence the learner writes in their own words.

Nobody is expected to know any of this beforehand. Every new word is explained with a picture and an everyday example, and the instructor works with each learner at their own pace.

How progress is checked

Every mark points to something the learner made. Marks are private, and learners are never ranked in public.

Six weekly portfolios	**20%**
Checkpoint one, end of week 4	**15%**
Checkpoint two, end of week 6	**20%**
Final project	**30%**
Individual demonstration	**15%**

To earn the certificate
At least 20 of the 24 classes attended or made up.
Both checkpoints, the final project and the demonstration handed in.
An overall mark of at least 50 out of 100.

Using AI tools

In weeks 1 to 6 learners write and fix their own code, so that the foundations are theirs. In weeks 7 and 8 they see how modern AI coding assistants work, and learn to check what such a tool says before trusting it. The final demonstration is done without AI help, and the certificate shows what the learner can do themselves.

What we ask of families

- Your child brings their own laptop and charger, Windows or Mac. We check it works with a short setup session before day one.
- Learners attend on time and tell us ahead of any absence, so a class can be made up.
- Parents and guardians come to the final demonstration in week 8.
- Photographs and sharing a project publicly are optional, and need a separate written consent.

Fee	UGX 1,000,000 per learner. The same for a group of one to five.
Venue	The National ICT Innovation Hub, Nakawa.
Schedule	Three classes a week for eight weeks, starting December 2026. Dates and times are confirmed at enrolment.
Time	24 classes of 3 hours, plus 4 labs of 2 hours: 80 scheduled hours including breaks.
Groups	No more than five learners in a group, so each one gets personal attention.

Your tutor. Jotham Martin Wambi, data scientist and ML/AI engineer, AI and ML lecturer at IUEA.

This is a beginner programme. It builds real skills and a project to show, and is not a professional qualification.